

DATABASES COURSE 1

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- Additional information: bd.ase.ro -> Databases, online.ase.ro

- About examination:
 - 50% seminar work (**30% project** - mandatory, 50% test, 20% activity)
 - 50% exam – multiple answers questions + SQL exercises
 - min. 5 at both of them
- About re-examination:
 - The re-examination exam can be taken, even if the seminar grade <5
- Oracle Academy on-line courses (to be finished by **16th of December 2024** + Oracle Academy **Final Exam** taken in week 12) -> bonus for the seminar

- Associate Professor Anda Belciu, PhD (groups 1072, 1076)
- Laurențiu Dincă, PhD-candidate (groups 1074, 1075)
- Alin Văduva, PhD-candidate (group 1073)

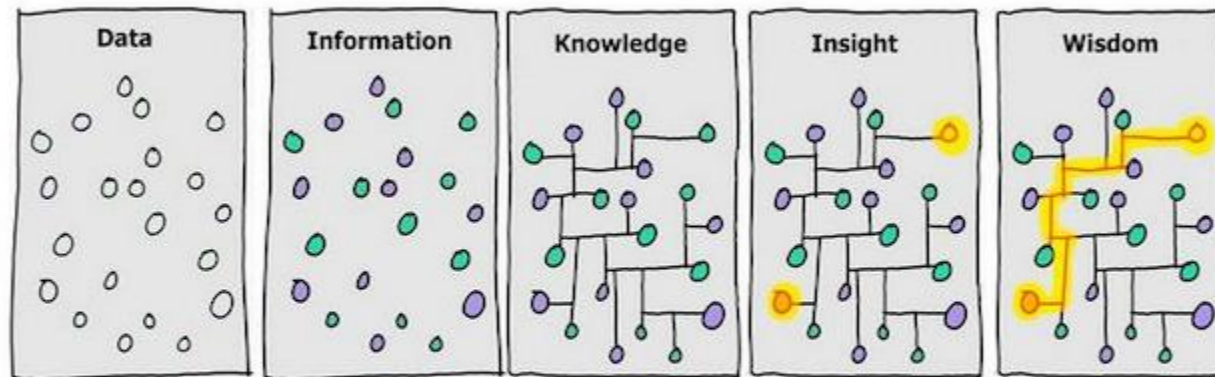
Databases – teaching staff

- C.J.Date , An introduction to database systems, Addison Wesley, 2004, USA.
- R.Elmasri, S.Navathe, Fundamentals of database systems, 7th edition, Pearson, 2016, USA.
- I. Lungu et al., Baze de date. Organizare, proiectare și implementare (Databases. Organization, design and implementation), ASE Publishing House, Second Edition, 2015, Romania.
- **A. Belciu, Introduction to Databases, ASE Publishing House, 2016, Romania.**

Bibliography

- DB theory
- Organizing data in external memory
- Relational databases
- SQL language
- Methodology for developing a database
- Indexing
- Clusters management in relational databases
- Queries optimization
- NoSQL DB and other types of databases

Course structure



[Cartoon by David Somerville, based on a two pane version by Hugh McLeod.]

Data

- Example:

Information

Employee Smith has 40 customers.

Knowledge

Employee Smith has 40 customers, while the other employees have an average of 20 customers.

Insight: 40 vs 20 customers

Wisdom

Employee Smith will have a salary raise.

- Volume – huge
- Velocity – high speed
- Variety – unstructured
- Variability – the meaning of data is constantly changing
- Veracity – uncertainty of data
- Visualisation – through charts & graphs
- Value – adds value to the business

- The 17 V's Of Big Data

BIG DATA

- **Databases – Definitions**

- Database (DB) = a set of data that is organized, coherent, structured, in terms of minimum redundancy and control, accessible to multiple users in a useful time.
- Database = a set of data collection that are in interdependence, along with the data description and the links between them.

Chapter I – Database theory

◦ Evolution

- In a computer, data is stored in the internal memory (temporarily) and in the external memory (persistent).
- In external memory, the way the data storage evolved has been established by:
 - access to data can be made as quickly and as easily as possible;
 - storing a large amount of data;
 - increasing data complexity;
 - improving the equipment for collecting, storing, transmitting and processing the data.
- Databases, seen as a way of organizing data in external memory, have evolved from files through a process of their integration (the files and links between them), and taking into account the related applications.

- Generations:
 - first = hierarchical and network DB (until the late 70s);
 - second = relational DB (late '70s until now);
 - third = object-oriented DB (late 80s and up to now).
- Advanced DB: spatial, MM, mobile, NoSQL (distributed DB, Cloud DB, Graph DB)
- Distributed DB vs Centralized DB
- Network DB (fixed record with a predefined set of fields) vs Graph DB (arbitrary key/value pairs on nodes/vertices and relations/edges).

CONTRACTS

ID	TYPE	VALUE
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DEPARTMENT

ID	NAME	MANAGER_ID
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PERSON

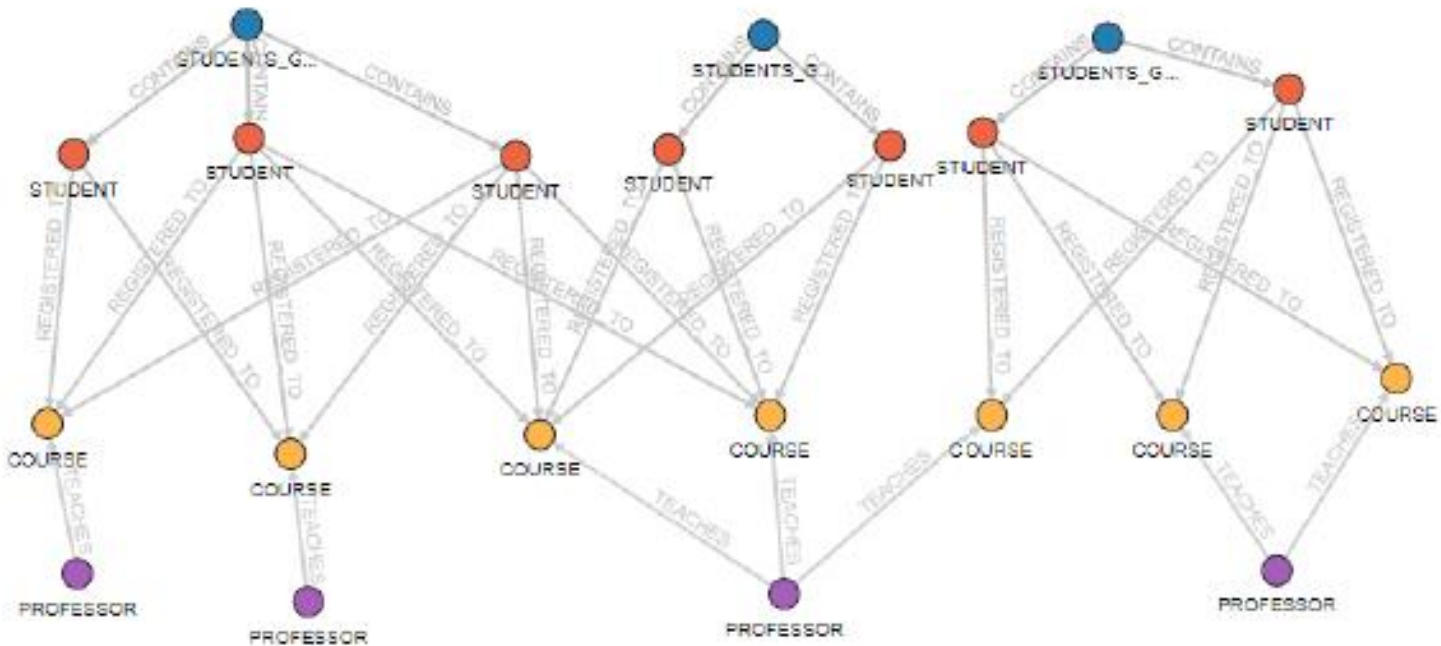
ID	NAME	ADDRESS
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CHILDREN

NAME	BIRTH
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- For this network structure a set type is: the owner node DEPARTMENT and PERSON as a member node.
- A realization of this set type can be:
- DEPARTMENT 100 IT 001
- PERSON 005 POPA ANDA MOȘILOR, 22
- PERSON 010 PETRE ION EMINESCU, 10

Network DB

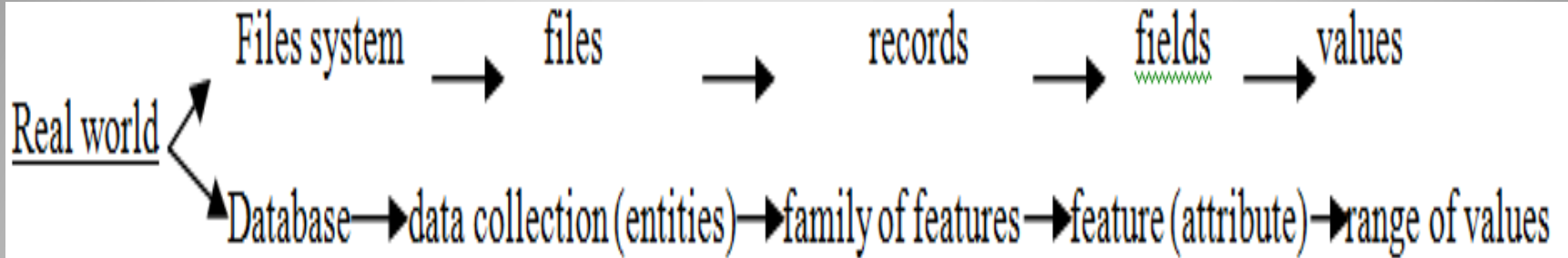


The correspondence between a row in Students table and the specific node (vertex) from the graph, which has properties and values indicated by standard key-value pairs.

Graph DB representation

Type of data organizing	Data structure	Data
Files	-> is kept in programs -> C example: <pre>File * f1; Struct employee { int id; char name[20]; ...};</pre>	-> is kept in the main file (on disk) -> extension example: .txt
DB	-> is kept in DB dictionary (on disk) through metadata -> Oracle example: <pre>create database db1; create table employees (id number (5), name varchar2(20)...);</pre>	-> is kept separately in a disk file -> extension example: .dbf

DB components



Range of values = the set of possible values for a feature (e.g. possible colors for a car).

Feature (attribute) = defining and describing a particular aspect (property) of a real-world entity (e.g. car brand).

Family of features = all characteristics that refer to the same real-world entity (e.g. all characteristics that can describe a car {number, brand, capacity, color}).

Data collection (entity) = a family of characteristics on which a predicate applies (leading to an order relation between the characteristics and to obtaining information with a certain meaning) which is affecting some connections.

Database vs. file system